

to the market? You should also think about how the market's psychological biases can change. Momentum strategies are negatively skewed; the skewness of the momentum strategy in Figure 7.7 is -1.43 . At a minimum, the investor should be able to tolerate large drawdowns induced by momentum strategies. Historically, these declines are concentrated in periods when policymakers have interrupted natural progressions of momentum, as in the Great Depression and the financial crisis.

4. Value Investing Redux

Factor risks represent bad times for an investor. There are two main types of factors—macro factors and investment factors. Assets are exposed to factor risks. The higher the exposure for a factor with a positive risk premium (the higher the asset's beta), the higher the asset's expected return.

The value strategy is an example of an investment style factor. In a rational story, value produces losses during bad times, and value stocks are risky. These bad times could coincide with bad times of the economy, as proxied by poor economic growth or poor returns of the market, or they could correspond with bad outcomes of other factors like firm investment. The average investor dislikes these bad times and requires a risk premium to hold value stocks. Thus, value stocks earn high returns to compensate investors for lousy returns during bad times. In behavioral stories, value stocks have high returns because investors underestimate the growth rates of value stocks. They overextrapolate the past growth rates of growth, or glamour, stocks, leading to growth stocks being overpriced and value stocks underpriced. If these behavioral biases are not arbitraged away, value stocks have high excess returns.

In the next two chapters, we turn to characterizing the factor risk–return trade-offs of the bread-and-butter asset classes, equities and fixed income, which can be considered factors in their own right.